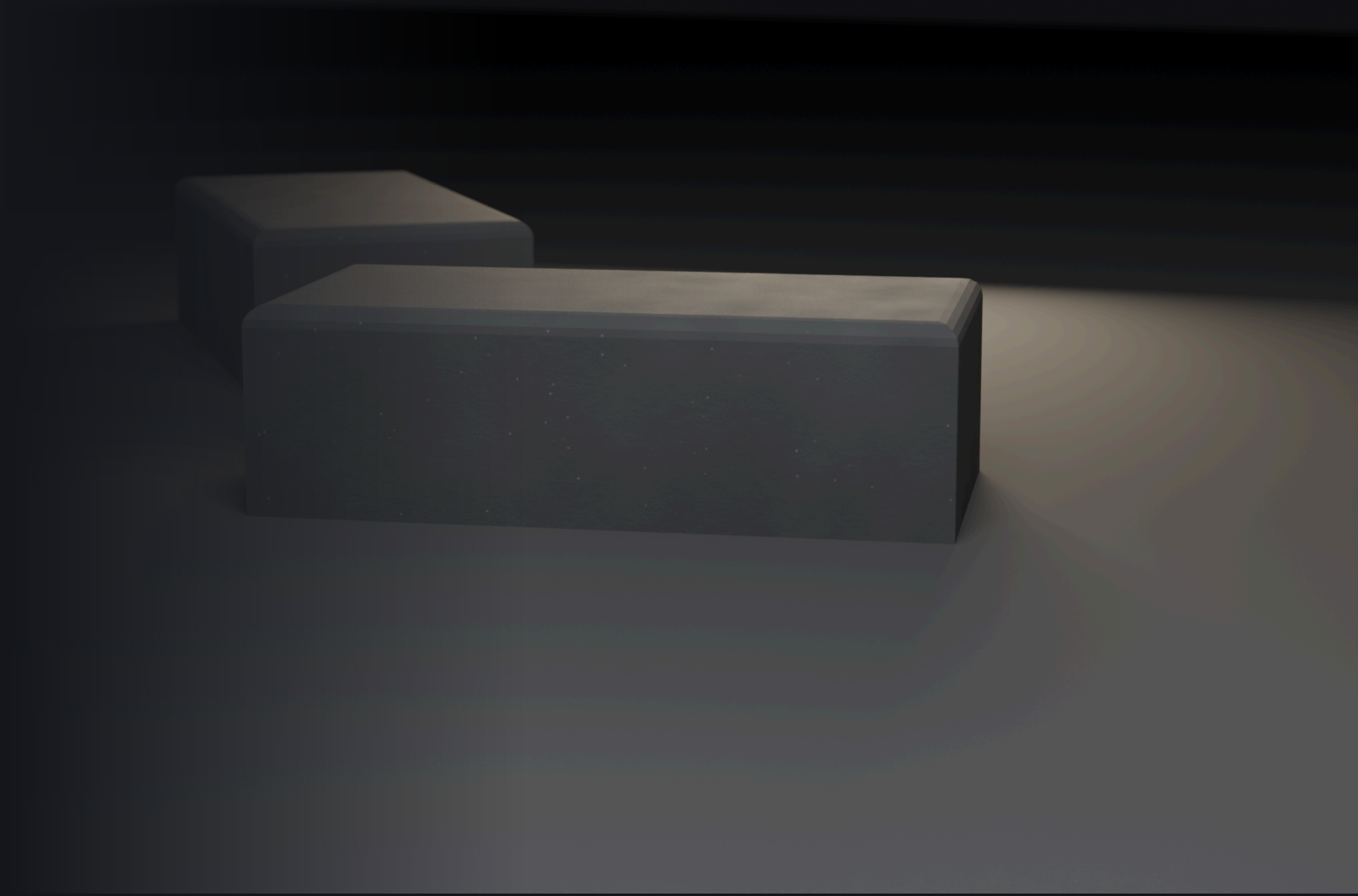
AMD DEVELOPER HACKATHON · ACT II · TRACK 3 UNICORN

The AI committee rejected the cheapest, greenest mix.

Because the bricks would have failed on the factory floor. That judgment — cheaper, greener, **but never below spec** — is MixMind.



Gemma 4 12B · LoRA fine-tuned | AMD Radeon W7900 + Instinct MI300X · ROCm | strength model R^2 0.910 · measured | open source

LIVE — hellotravisss.github.io/mixmind/web

MixMind — a plant-private concrete-mix AI on AMD   github.com/Hellotravisss/mixmind   hellotravisss.github.io/mixmind/web

THE PROBLEM

Concrete is 8% of global CO_2. The know-how to fix a mix sits in labs most plants can't afford.

And no plant will ever upload its recipes to a third-party AI — the mix design **is** the business.

~8%	1 in 1000s	0
of global CO_2 is cement & concrete	plants has a mix-R&D department	recipes a plant will send to a cloud API

So the intelligence has to run **inside the fence** — on hardware the plant owns.

MIXMIND

A plant's R&D department, self-hosted on its own AMD GPU.

Floor Copilot `RAG · cited`

Answers the plant's own shop-floor knowledge — silo bridging, curled edges, wet sand — cites the note id, and refuses when it doesn't know.

Mix Committee `4 agents · grounded`

Standard, QC, Cost and Carbon officers debate a recipe change against a real strength model, and stamp a verdict.

Facts live in a **swappable knowledge base** — point the same fine-tuned model at another plant's notes and it serves that plant. The recipe never leaves the building.

> Material is stuck in the cement silo — what do I do?
>
> Give a few firm hits near the outlet with a rubber mallet to break the bridge KB-07. If it still won't drop, lock out the feed first, then open the inspection cover near the base KB-07.

> What is the current market price of grey cement?
>
> I don't have a note on that.

100%	0
grounded citation accuracy — 9/9 held-out phrasings, correct note cited, distractor note avoided	invented answers — off-notes questions get an honest refusal, by training and by design

MIX COMMITTEE · FOUR SPECIALISTS, REAL NUMBERS

Every claim in the debate is a computed number.

PROPOSED CHANGE	28-DAY MPA	2-DAY · DEMOULD ≥12	COST/UNIT	CO_2	VERDICT
Sweet spot	52.6	16.1 ✓	$87.9	−17%	APPROVED
20% cement → slag	46.3	12.7 ✓ tight	$88.2	−18%	CONDITIONS · 1-week pilot
Aggressive · 45% slag	36.0	8.0 ✗	$84.8	−40%	REJECTED

Strength: gradient boosting on **1,030 real lab tests** (UCI · Yeh 1998), held-out **R^2=0.910, MAE 3.46 MPa**. Cost & CO_2: deterministic arithmetic. The LLM writes the debate — **never the numbers**.

THE MOMENT THAT MATTERS

REJECTED

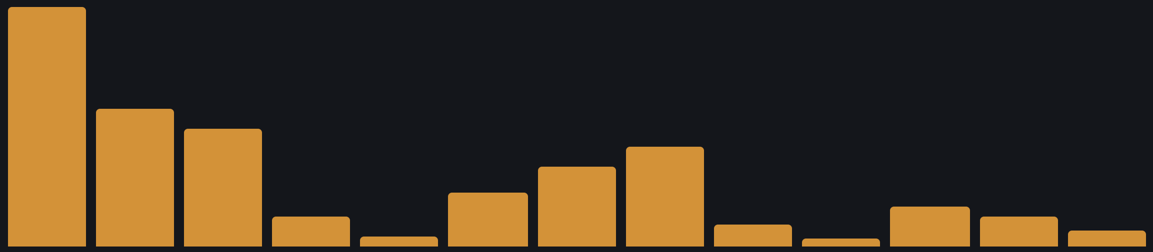
The 45%-slag mix saves the most money and cuts CO_2 40% — and the committee kills it.

Predicted 2-day strength: **8.0 MPa**, below the 12.0 demould spec — the units would stick and chip at demould. The model learned slag's early-strength penalty **from the data**; nobody hard-coded that rule. Judgment, not a yes-man.

FINE-TUNED ON AMD · MEASURED, NOT PROMISED

A 287 MB adapter turns Gemma 4 into this plant's engineer.

```
LoRA r=16 · 65.5M trainable (0.55% of 12B)
42-example plant SFT set · 3 epochs · 60 steps
loss 3.029 → 0.001 · <1 min on MI300X
saved adapter → mixmind-gemma4-lora (287 MB)
```



training loss · 60 steps

Measured before → after on held-out prompts:

diagnosis	answers now **lead with the likely culprit** instead of paraphrasing the note
−15% length	tighter, plant-voice answers
judge 2/3	independent **Fireworks LLM-judge** (Kimi K2, blind A/B) preferred the fine-tuned answer on 2 of 3 held-out prompts, tie on the third
judged	**Fireworks LLM-judge** (blind A/B, independent model family): preferred the fine-tuned answer **2 of 3** held-out prompts
honest	citations were already 100% with prompting — fine-tuning bakes the **behavior into the weights**, and each plant retrains on its own floor data

WHY AMD · THE PART EVERYONE ELSE ARGUES DIFFERENTLY

Everyone proves AMD can go big. We prove it goes small enough to afford.

Radeon PRO W7900 · 48 GB `~$4k · in the plant`

GPU[0] Card Vendor: Advanced Micro Devices, Inc. [AMD/ATI] · gfx1100
VRAM Total: 51,522,830,336 B

Runs the full product — fine-tuned Gemma 4, Copilot, Committee. Private AI a small plant can actually buy.

Instinct MI300X · 192 GB `$1.99/hr · for training`

GPU[0] Card Series: AMD Instinct MI300X VF · gfx942
VRAM Total: 205,822,885,888 B

Fine-tune in under a minute on AMD Developer Cloud, ship the 287 MB adapter back to the plant card.

Train big, serve cheap — **both ends AMD**, verified with rocm-smi on real hardware (evidence in repo).

WHAT BROKE ON THE WAY · REAL TIME ON REAL HARDWARE

The failures we hit — and shipped through.

env	**!pip installs missed the Jupyter kernel** on the AMD notebook — deps must go in with %pip, then restart
versions	Gemma 4 needs **transformers ≥5.13**, which conflicts with the preinstalled ROCm vLLM (<5) — we serve via transformers, vLLM serving is a next step
tokenizer	transformers 5.x **apply_chat_template dtype crash** in our trainer — fixed with return_dict=True (train_lora.py)
model	with no notes provided, base **and** fine-tuned model can hallucinate a citation — v2 dataset adds "no notes → no citation" examples
honesty	citation accuracy was already 100% at base with prompting — we report it as a **system** metric, not a fine-tune gain

HONEST BOUNDARY

We predict directions.
We don't certify mixes.

The strength model is trained on wet-cast lab data; dry-cast paver production differs. So MixMind proposes and screens — the plant's QC still owns the final call, and load-bearing changes go to a qualified engineer. **That division of labor is the product, not a disclaimer.**

THE BUSINESS · BOTTOM-UP, ALL FIGURES PUBLIC-BALLPARK ESTIMATES

A mid-size plant spends ~$1M+ a year on cement. We sell the tool that trims it.

Mid-size paver/block plant (~30 t cement/day) → **~$1.0–1.5M/yr** cement spend

SCM substitution + trimming over-design → **5–19% less cement** committee-screened, spec held

Recovered per plant → **$50K–250K / plant-yr** + proportional CO_2 cut

MixMind at **$2K/plant/mo**, self-hosted on one AMD GPU → **<20% of savings** pays back in the first month

The alternative is a materials engineer (**$90–120K/yr**) or a consultant retainer — which is why most of the ~3,000 North-American precast plants simply don't optimize. That's a **$0.3–0.8B/yr** recoverable-spend wedge, and the on-prem AMD box is what makes the price point possible.

WHERE THIS GOES

Every plant that can't afford a research lab gets one it can run itself.

1	swap the KB	tonnes
GPU in the plant office — the whole stack, on-prem	same model serves the next plant — knowledge is retrieval, not weights	of CO_2 per plant-year from cement cut at held strength

Small precast plants, block producers, builders in underserved regions — the operators who touch the mix every day, finally with an R&D department that answers back. **Built by one of them.**

TRY IT YOURSELF · 90 SECONDS, NO LOGIN

hellotravisss.github.io/mixmind/web

1. **Ask the floor:** type a real shop-floor question — or try "what's the market price of cement?" and watch it **refuse** instead of guess.
2. **The committee:** pick **"Aggressive · 45% slag"** — the cheapest, greenest mix — and watch QC stamp it **REJECTED** on early strength.
3. **Drive the mix:** drag cement down until the verdict flips; type **your** cement price into the price panel and watch the economics recompute.

−17%	R^2 0.910	100%	0 bytes
CO_2, strength held	strength model · 1,030 lab tests	grounded citations · 9/9 held-out	of recipe data sent off-site

White cement, grey cement, slag, chips — I touch these every day.

MixMind is a machine operator's answer to a question the industry keeps asking: what if the plants that make the world's most-used material could afford the intelligence to make it cleaner?

Open the live demo. Ask it a floor question. Pull the cement down.